Algebra · architecture

The router and spend firewall for AI agents that pay for APIs on Solana

You (the person)
Sign in with Privy
(Solana wallet or email code)
Create Spend Passes
Approve · freeze · kill switch

Web console
Next.js
Passes · firewall · routing
Providers · executions
Agent chat

AI agents
1
Claude Code · Cursor
ChatGPT agents · any bot
Algebra console chat
Hold a Spend Pass token,
never a wallet key

REST API
/api/v1/execute · simulate

MCP server
/mcp · 7 tools

Owner's Solana wallet
Phantom · Solflare · Backpack
Funds the pass vault
Freezes it on chain

fund · freeze

Algebra backend
Go · one process · REST and MCP call the same services

Spend firewall
2
A Spend Pass per agent
Budget · per-call cap
Approval line · allow-list
Calls per minute
Rule for never-paid providers
Kill switch · dry run

Router
3 4
14 classes of work
Free unpaid 402 quotes
Guards drop down, overpriced
and trap-priced providers
Rank: cheapest · fastest · auto
Falls back if nothing was paid

Economic coordinator
5
intent → reserve → begin → authorize → complete → reconcile
One live attempt per outcome, enforced by Postgres: never pays twice

Receipts and replay
8
Ed25519-signed receipt
Anyone verifies: /verify, JWKS
Answer kept sealed for 24 h

Payment rails
6 7
x402 exact (USDC)
x402 upto (metered)
Jupiter swap (off by default)
Sandbox (demo, no real money)
Ceiling: 1 USDC a payment
The key never leaves the rail

PostgreSQL · Redis
Postgres is the source of truth
Redis: rate limits only

Discovery
1,622 providers in 4 catalogs, plus the open web

- Pay.sh · 75 providers
- Circle Agent Marketplace · 27
- PayAI bazaar · 1,152
- Coinbase x402 Bazaar · 368
- Gemini web search · unlisted APIs

Paid API providers (x402)
4 7
Token price · token risk · search · LLM …
Ask: HTTP 402 with a price
Pay: a signed USDC transfer
Answer: treated as untrusted data

pull

USDC to provider

Solana
USDC · mainnet and devnet kept apart

Spend Pass program (Anchor)
6
devnet · 46gQ…9tdS
A vault per pass, owner-funded
Limits checked on chain
Pulls the exact amount first
Unused pulls refunded

Algebra's payer wallets
7 8
One per cluster
Sign x402 USDC transfers
Facilitator settles them
Settlement proven from chain

Other Solana programs
Payment channels (x402 upto):
escrow, settle, refund
Jupiter: token swaps
SPL Token: USDC

One paid request

1. Agent asks: work, input, max price (REST or MCP)
2. Spend firewall checks the pass: allow, ask you, deny
3. Router gathers candidates: catalogs by class, web
4. Free 402 quotes; guards drop bad ones; rank
5. Coordinator reserves one attempt (never twice)
6. On-chain pass: program pulls exact amount
7. Rail pays the provider in USDC; provider answers
8. Settlement proven from chain; signed receipt back

Give agents permission to spend, not access to money.

Demo accounts pay only in the sandbox. Every limit is checked again right before the wallet signs.

github.com/anshitraj/algebra-ai